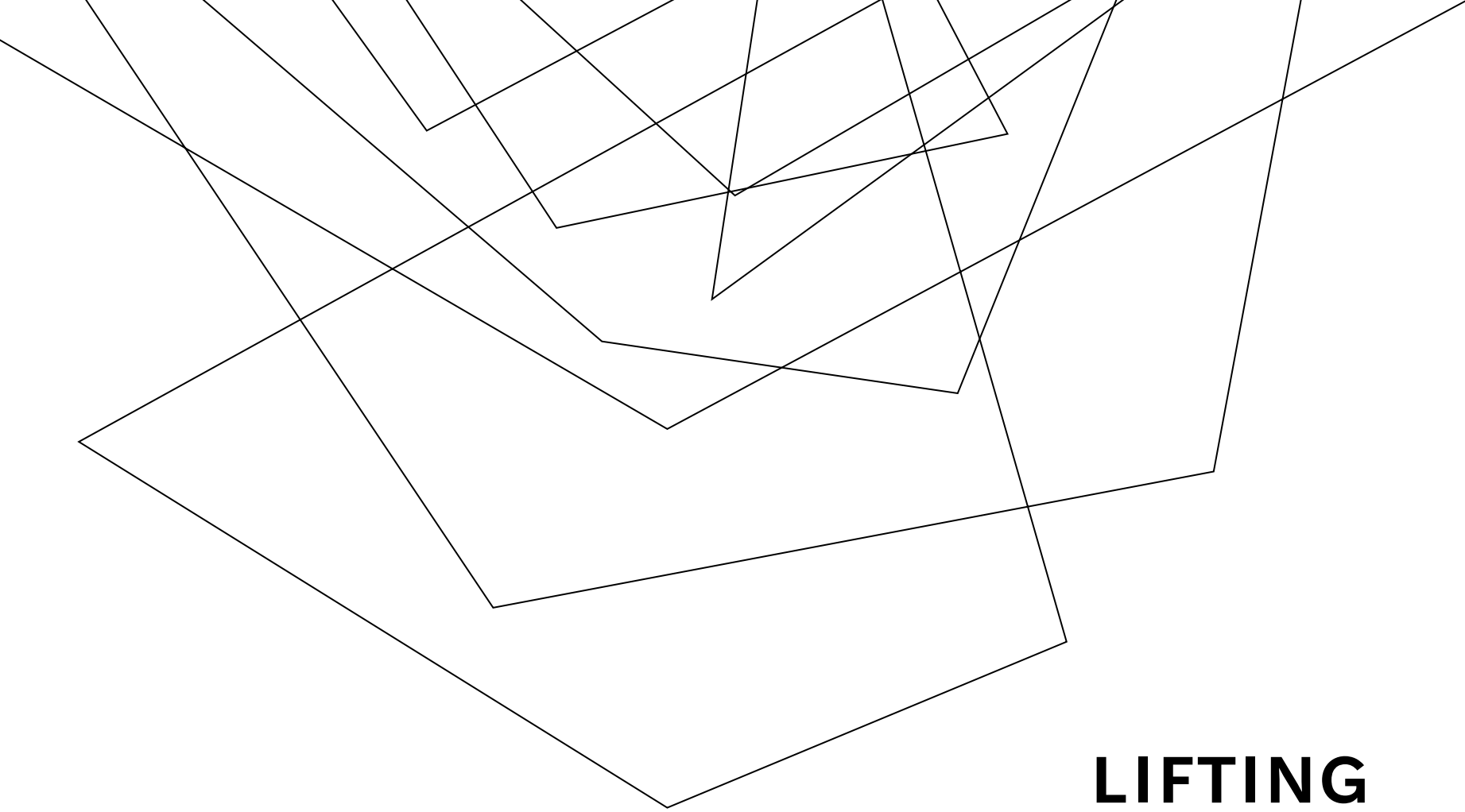




LIFTING
an overview



"PROPER LIFTING TECHNIQUES
ARE ESSENTIAL TO PREVENT
INJURIES, ESPECIALLY WHEN
DEALING WITH HEAVY
WEIGHTS." – UNKNOWN
ERGONOMIST



"WHEN LIFTING HEAVY
OBJECTS, ALWAYS REMEMBER
TO BEND YOUR KNEES AND
KEEP YOUR BACK STRAIGHT TO
AVOID INJURY."
– DR. JOHN SMITH,
ERGONOMICS SPECIALIST



"WHEN LIFTING HEAVY
OBJECTS, IT'S CRUCIAL TO USE
PROPER BODY MECHANICS TO
AVOID INJURY."
– DR. KAREN JACOBS,
PROFESSOR OF OCCUPATIONAL
THERAPY

MY QUESTIONS GOING IN

Standard Orgs and
Guidelines

Material Handling Best
Practices

Practical Thresholds



STANDARDS ORGS AND GUIDELINES



OSHA

Occupational Safety and Health Administration

- enforcement arm
- has no hard and fast lifting rules
- provides industry specific material pamphlets and solicits reports of bad practice.

OSHA GUIDELINES EXAMPLE

Taken from OSHA ergonomic principals for Electrical Contractors

- ...Sometimes it is appropriate to lift with the legs, and other times the back should be used to lift. These techniques depend on the size and shape of the load, and the frequency of lifting that is required.
- Proper handholds make lifting easier and reduce the risk of injury. Handholds should be made large enough to accommodate larger hands and should not dig into fingers and palms.
- Rotate to jobs that affect different parts of the body or have different intensities.



ISO

International Standards Organization

- Go-to guys for standards
- you have to pay to see their standards
- 11228-1 lifting lowering and carrying



NOISH

National Institute for Occupational Safety and Health

- Research arm that feeds OSHA with information, part of the CDC
- Has research and guidance about avoiding WRMDs (Work-Related Musculoskeletal Disorders) more general information with the audience being someone in charge of a factory etc.
- Lifting equation ([calculator](#))

NOISH LIFTING EQUATION

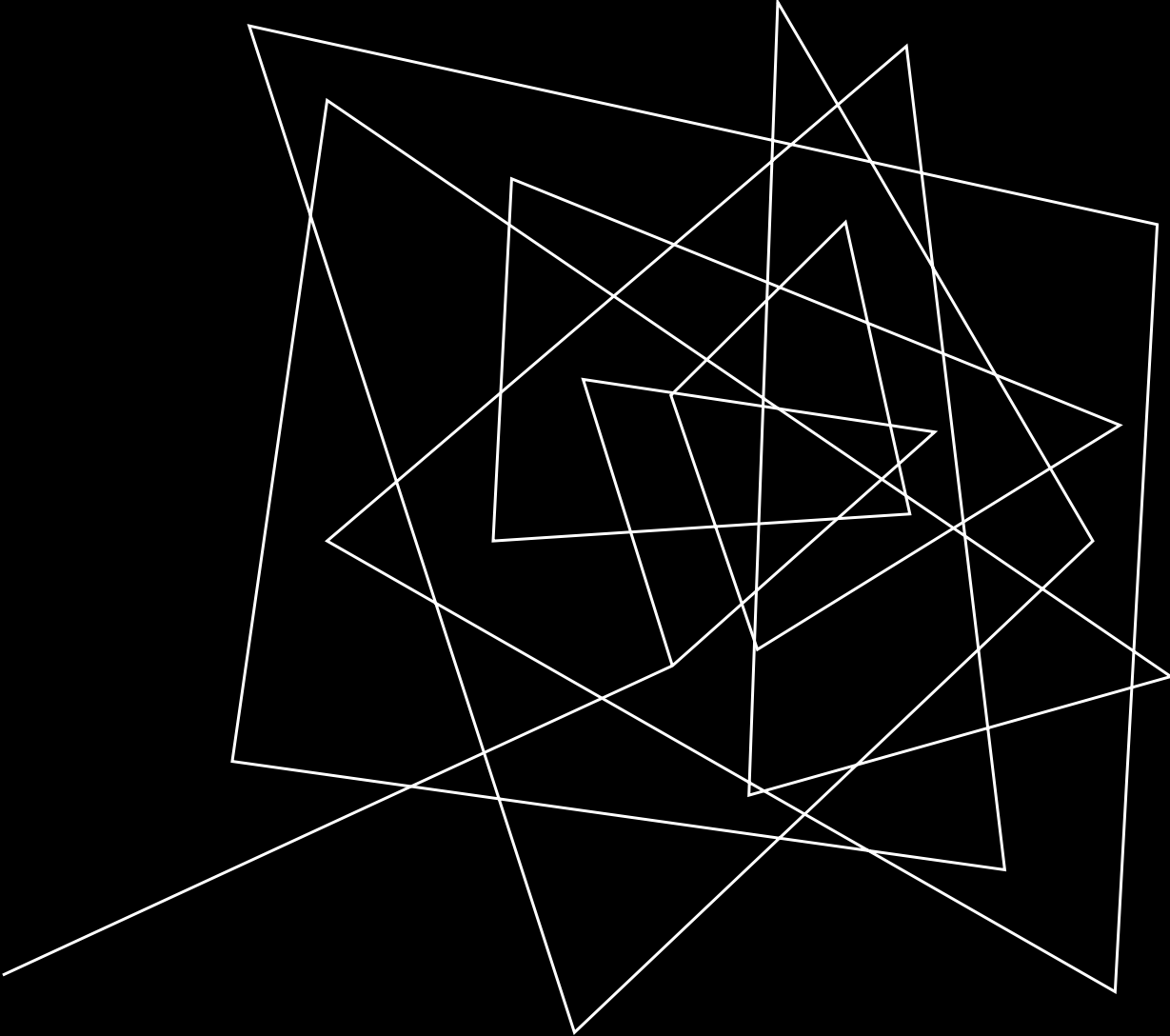
Recommended Weight Limit (RWL)

$$RWL = LC * HM * VM * DM * AM * FM * CM$$

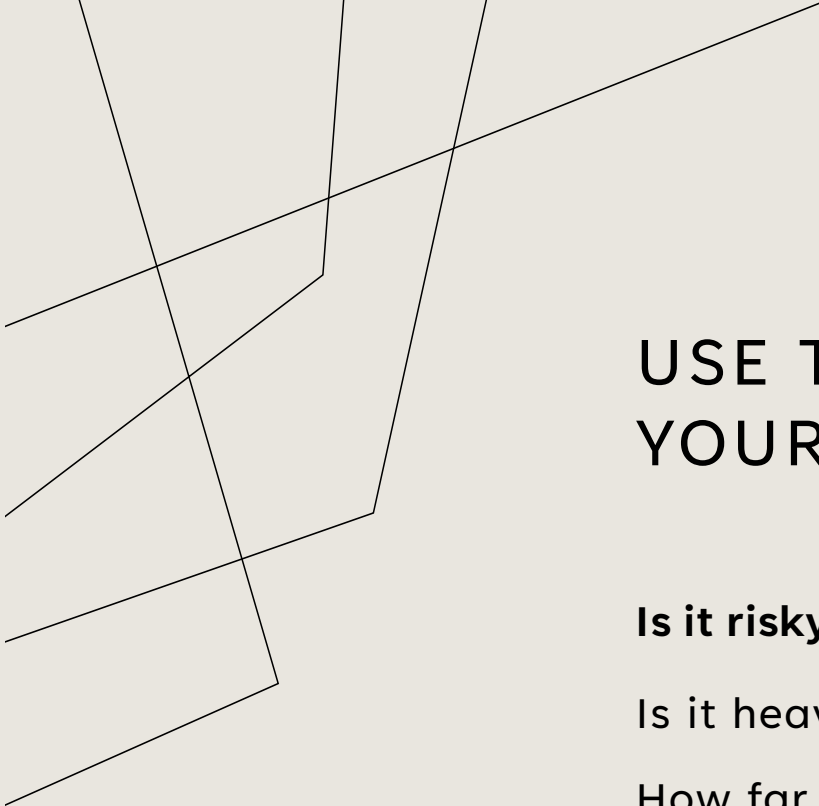
Lifting Index

$$LI = \frac{\text{Load Weight}}{\text{Reccomended Weight Limit}} = \frac{L}{RWL}$$

		Metric	U.S. Customary
Load Constant	LC	23kg	51lb
Horizontal Multiplier	HM	(25/H)	(10/H)
Vertical Multiplier	VM	1 – (.003 V-75)	1 – (.0075 V-30)
Distance Multiplier	DM	.82 + (4.5/D)	.82 + (1.8/D)
Asymmetric Multiplier	AM	1 – (.0032A)	1 – (.0032A)
Frequency Multiplier	FM	From Table 5	From Table 5
Coupling Multiplier	CM	From Table 7	From Table 7



MATERIAL HANDLING BEST PRACTICES



USE THE NOISH EQUATION AND WORK IT IN YOUR MIND BACKWARDS

Is it risky?

Is it heavy?

How far away do I have to reach?

How high do I have to lift it?

How far do I need to move it?

How much do I have to twist?

How often do I do it?

How good is my grip?

How to derisk

Make it lighter

Hold it closer

Don't lift it so high

Don't move it far

Don't require a twist

Don't do it often

Use good handles

SYNTHESIZED WEIGHT
THRESHOLDS ARE AS FOLLOWS

<3KG – LITTLE TO NO RISK OF
INJURY

3KG-10KG – PROCEED WITH
CAUTION, TASK CAN BE
DESIGNED TO DERISK

>10 KG - DANGEROUS